



Figure 5: Preprocessing – Open data set

inclusion and exclusion of attributes from the experiment.

- 6) **Visualize:** This allows us to see the possible visualisation produced on the data set in a 2D format, in scatter plot and bar graph output.

The user cannot move between the different tabs until the initial preprocessing of the data set has been completed.

**Preprocessing:** Data preprocessing is a must. There are three ways to inject the data for preprocessing:

- **Open File** – enables the user to select the file from the local machine
- **Open URL** – enables the user to select the data file from different locations
- **Open Database** – enables users to retrieve a data file from a database source

A screen for selecting a file from the local machine to be preprocessed is shown in Figure 5.

After loading the data in Explorer, we can refine the data by selecting different options. We can also select or remove the attributes as per our need and even apply filters on data to refine the result.

**Classification:** To predict nominal or numeric quantities, we have classifiers in Weka. Available learning schemes are decision-trees and lists, support vector machines, instance-based classifiers, logistic regression and Bayes' nets. Once the data has been loaded, all the tabs are enabled. Based on the requirements and by trial and error, we can find out the most suitable algorithm to produce an easily understandable representation of data.

Before running any classification algorithm, we need to set test options. Available test options are listed below.

**Use training set:** Evaluation is based on how well it can predict the class of the instances it was trained on.

**Supplied training set:** Evaluation is based on how well it can predict the class of a set of instances loaded from a file.

**Cross-validation:** Evaluation is based on cross-validation by using the number of folds entered in the 'Folds' text field.

**Split percentage:** Evaluation is based on how well it can

predict a certain percentage of the data, held out for testing by using the values entered in the '%' field.

To classify the data set based on the characteristics of attributes, Weka uses classifiers.

**Clustering:** The cluster tab enables the user to identify similarities or groups of occurrences within the data set. Clustering can provide data for the user to analyse. The training set, percentage split, supplied test set and classes are used for clustering, for which the user can ignore some attributes from the data set, based on the requirements. Available clustering schemes in Weka are k-Means, EM, Cobweb, X-means and FarthestFirst.

**Association:** The only available scheme for association in Weka is the Apriori algorithm. It identifies statistical dependencies between clusters of attributes, and only works with discrete data. The Apriori algorithm computes all the rules having minimum support and exceeding a given confidence level.

**Attribute selection:** Attribute selection crawls through all possible combinations of attributes in the data to decide which of these will best fit the desired calculation—which subset of attributes works best for prediction. The attribute selection method contains two parts.

- **Search method:** Best-first, forward selection, random, exhaustive, genetic algorithm, ranking algorithm
- **Evaluation method:** Correlation-based, wrapper, information gain, chi-squared

All the available attributes are used in the evaluation of the data set by default. But it enables users to exclude some of them if they want to.

**Visualisation:** The user can see the final piece of the puzzle, derived throughout the process. It allows users to visualise a 2D representation of data, and is used to determine the difficulty of the learning problem. We can visualise single attributes (1D) and pairs of attributes (2D), and rotate 3D visualisations in Weka. It has the Jitter option to deal with nominal attributes and to detect 'hidden' data points. 

## References

- [1] Weka Machine Learning Project, <http://www.cs.waikato.ac.nz/~ml/index.html>
- [2] E. Frank, Machine Learning With Weka, University of Waikato, New Zealand
- [3] B. Mobasher, Data Preparation and Mining with Weka, [http://maya.cs.depaul.edu/~classes/ect584/WEKA/association\\_rules.html](http://maya.cs.depaul.edu/~classes/ect584/WEKA/association_rules.html), DePaul University, 2003

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